Cessna 210R Centurion (C210R)

Continental IO-520-L, 300 hp · 6 seats · Typical 210R (1985–86, IO-520-L, 4,100 lb takeoff / 3,800 lb landing, 38 ft 10 in wing, unpressurised) POH procedures: electric hydraulic power pack with NO main gear doors (wheels retract into open wells and are held up by hydraulic pressure), hand-pump emergency extension, electric pre-select flaps, LEFT/RIGHT/OFF fuel selector with vapour return to the selected tank; the door-equipped 210M and the turbocharged T210R / P210R differ.

V-speeds & limits

Vr	**65–70 KIAS (liftoff, flaps 0–10°)**
Vx	**72 KIAS (sea level, flaps up)**
Vy	**95 KIAS (sea level, flaps up)**
Best glide	**85–90 KIAS (4,100 lb, gear & flaps up, prop low RPM)**
Vfe	**160 KIAS (10°) / 140 KIAS (20°) / 120–140 KIAS (30° FULL – verify placard; 210R checklists differ)**
Vlo	**140 KIAS (extend & retract)**
Vle	**165 KIAS (extended; verify placard)**
Va	**approx 133 KIAS at 4,100 lb (lower when lighter, approx 120 KIAS at 3,300 lb)**
Vno	**167 KIAS**
Vne	**204 KIAS**
Vs / Vso	**approx 66 / 59 KIAS (4,100 lb)**
Vref	**75–80 KIAS final (flaps 30°); 70–75 KIAS short field; 85–90 KIAS flaps up**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel sumps drained, oil quantity, open main gear wells clear – no gear doors on the R model, hydraulic fluid level checked)**
Passenger briefing	**COMPLETE (belts, both cabin doors, exits, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin doors (both) / baggage door	**CLOSED & LATCHED**
Parking brake	**SET**
Avionics power switch	**OFF**
Electrical equipment	**OFF**
Circuit breakers	**CHECK IN (including landing gear pump)**
Landing gear lever	**DOWN**
Fuel selector	**LEFT or RIGHT – fullest tank (there is no BOTH; feel for the detent)**
Fuel quantity	**CHECK both gauges against visual check (approx 87–90 gal usable total – verify POH)**
Cowl flaps	**OPEN**
Alternate static source	**OFF (normal)**
Master switch (BAT & ALT)	**ON**
Gear-down light (green)	**ON (press to test gear-up light)**
Low-voltage light	**OUT after start (may be on with engine stopped)**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR**
Propeller	**HIGH RPM (full forward)**
Throttle	**OPEN approx 1/4 to 1/2 in**
Mixture	**IDLE CUTOFF**

▶ Engine hot or vapour-locked?

No – normal prime:

Auxiliary fuel pump	**ON**
Mixture	**FULL RICH until fuel flow indicates approx 6–8 GPH (3–5 sec), then IDLE CUTOFF**
Auxiliary fuel pump	**OFF**

Yes – hot start:

Throttle	**FULL OPEN**
Auxiliary fuel pump	**ON approx 20–30 sec with mixture IDLE CUTOFF to purge vapour from the lines, then OFF**
Throttle	**1/4 to 1/2 in OPEN**
Mixture	**FULL RICH for approx 2 sec, then IDLE CUTOFF**

If the engine floods: mixture IDLE CUTOFF, throttle FULL OPEN, crank; retard throttle and advance mixture as it fires.

Ignition switch	**START (release when engine fires)**
Mixture	**ADVANCE SMOOTHLY to FULL RICH as engine fires**
Throttle	**800–1,000 RPM**
Oil pressure	**10 psi minimum within 30 sec (normal 30–60 psi)**
Ammeter / low-voltage light	**CHARGING / OUT**
Avionics power switch	**ON**
Hydraulic pump	**NOTE – may run briefly to build pressure, must stop**
Engine instruments	**CHECK**

Before Taxi

Brakes	**CHECK on initial roll**
Nosewheel steering	**CHECK**
Flight instruments	**CHECK & SET (altimeter, DG / HSI, turn coordinator and compass during turns)**
Avionics / GPS / radios	**SET**
Autopilot / electric trim	**CHECK & OFF**
Elevator & rudder trim	**SET for takeoff**
Fuel selector	**CHECK fullest tank (LEFT or RIGHT)**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Cabin doors / windows	**CLOSED & LATCHED**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH (lean for smooth run-up above approx 3,000 ft elevation)**
Throttle	**1,700 RPM**
Magnetos	**CHECK L & R (150 RPM max drop, 50 RPM max difference)**
Propeller	**CYCLE 2–3 times (RPM drops approx 300–500, then returns), then HIGH RPM**
Engine instruments	**CHECK (oil pressure 30–60 psi, oil temp in green, fuel flow steady)**
Ammeter / low-voltage light	**CHARGING / OUT**
Suction	**4.5–5.5 in Hg**
Throttle	**IDLE check (approx 600–650 RPM), then 1,000 RPM**
Fuel selector	**FULLEST TANK (LEFT or RIGHT – feel for detent)**
Auxiliary fuel pump	**OFF**
Cowl flaps	**OPEN**
Propeller	**HIGH RPM**
Flaps	**0°–10° (pre-select lever, check indicator; 10° for short field)**
Trim	**SET takeoff**
Landing gear lever	**DOWN – green light ON**
Pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (full throttle 2,850 RPM; 4,100 lb max takeoff, 3,800 lb max landing weight; abort point; gear up only with positive rate and no usable runway)**

After Takeoff / Climb

Brakes	**APPLY momentarily to stop wheel rotation**
Landing gear	**UP (positive rate, no usable runway, below 140 KIAS)**
Gear-up light	**ON, hydraulic pump stopped (no main gear doors – wheels sit in the open wells, held up by pressure)**
Flaps	**UP (above 70–75 KIAS, after obstacles)**
Climb power	**25 in Hg / 2,550 RPM cruise climb (full throttle / 2,700 RPM max continuous; 2,850 RPM takeoff limited to 5 min)**
Mixture	**LEAN per POH climb fuel-flow schedule above approx 5,000 ft**
Cowl flaps	**OPEN (CHT 460°F max, oil temp 240°F max)**
Auxiliary fuel pump	**OFF (ON briefly if fuel flow fluctuates)**
Airspeed	**110–120 KIAS cruise climb (95 KIAS Vy)**
Engine instruments	**CHECK**

Cruise

Cruise power	**SET (2,300–2,500 RPM, 20–25 in Hg typical, 75% max)**
Mixture	**LEAN per POH cruise table / EGT (approx 50°F rich of peak)**
Cowl flaps	**CLOSED (as required for CHT)**
Fuel selector	**ALTERNATE LEFT / RIGHT every 30–60 min – note the time, keep tanks within approx 10 gal**
Fuel quantity / flow	**CHECK against plan (vapour and excess fuel return to the SELECTED tank – do not run a tank to empty)**
Engine instruments	**CHECK (oil pressure 30–60 psi, oil temp 240°F max)**
Hydraulic pump	**BRIEF RUNS NORMAL (gear is held up by hydraulic pressure, no uplocks); continuous running is abnormal**
Altimeter / heading	**CHECK**

Descent

Altimeter	**SET**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Mixture	**ENRICH progressively**
Power	**REDUCE gradually (avoid rapid cooling, keep CHT in green)**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 140 KIAS before gear extension**
Approach briefing	**COMPLETE**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Auxiliary fuel pump	**OFF**
Mixture	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 140 KIAS**
Landing gear lever	**DOWN – green light ON, pump stopped (no door cycle to hear on the R model – allow approx 10 sec)**
Flaps	**10° below 160 KIAS; 20° below 140 KIAS; FULL (30°) on final below 120–140 KIAS (per placard)**
Propeller	**HIGH RPM (on final)**
Airspeed	**75–80 KIAS final (70–75 KIAS short field)**
Landing light	**ON**
Landing gear	**GREEN – CONFIRMED on final, horn silent at idle**

After Landing

Flaps	**UP**
Cowl flaps	**OPEN**
Auxiliary fuel pump	**OFF**
Pitot heat	**OFF**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trim	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttle	**1,000 RPM (cool-down approx 1 min)**
Avionics power switch	**OFF**
Electrical equipment / lights	**OFF**
Mixture	**IDLE CUTOFF**
Ignition switch	**OFF (after propeller stops), key removed**
Master switch (BAT & ALT)	**OFF**
Fuel selector	**LEFT or RIGHT (leave on a tank, not OFF, to avoid cross-feed confusion; follow operator policy)**
Control lock	**INSTALL**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

Low-voltage light on, ammeter discharging (28 V system)

Avionics power switch ..... **OFF**

Alternator circuit breaker ..... **CHECK IN / RESET once**

Master switch ..... **OFF, then ON (both halves)**

▶ Low-voltage light out, ammeter charging?

Yes:

- Avionics power switch ..... **ON**
- Ammeter ..... **MONITOR**

No:

- Master switch ..... **ALT half OFF (BAT ON)**
- Non-essential electrical load ..... **REDUCE (battery only – pitot heat, lights, extra radios OFF)**
- Landing gear ..... **PLAN early extension while battery power remains (hand pump if the pump will not run)**
- Flaps ..... **PLAN to extend early or land flaps-up**
- Deferred: Alternator failed – battery only; extend gear early (hand pump if needed), land as soon as practical. ..... **NOTED**

Low Oil Pressure / High Oil Temperature

Oil pressure below 30 psi in cruise or oil temperature above 240°F

Oil pressure / oil temperature ..... **CHECK and cross-check**

Cowl flaps ..... **OPEN**

Mixture ..... **ENRICH**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

- Power ..... **REDUCE to minimum required**
- Airport ..... **DIVERT to nearest suitable**
- Engine ..... **EXPECT FAILURE – stay within glide of a landing area, plan gear extension early**
- Deferred: Low oil pressure / high oil temperature – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure / temperature indication suspect – maintenance check required. ..... **NOTED**

Fuel Flow Fluctuation / Vapour / Rough Running

Auxiliary fuel pump ..... **ON until fuel flow is steady (approx 1–2 min), then OFF**

Fuel selector ..... **SWITCH TANKS (fullest tank; feel for detent)**

Mixture ..... **ADJUST for smooth running**

Magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (fuel flow, CHT, EGT, fuel quantity on selected tank)**

▶ Engine running smoothly, fuel flow steady?

Yes:

- Deferred: Fuel flow / rough running cleared – leave aux pump OFF unless needed again; report to maintenance. ..... **NOTED**

No:

- Auxiliary fuel pump ..... **ON (leave on)**
- Power ..... **REDUCE to smoothest setting**
- Deferred: Engine rough / fuel flow unsteady – land as soon as practical at nearest suitable airport. ..... **NOTED**

Hydraulic Pump Runs Continuously

Pump heard running more than approx 1 min after a gear cycle, or running in cruise

Landing gear lever ..... **CHECK in the detent for the current gear position**

Gear lights ..... **CHECK (gear up with no gear-up light = gear not fully up)**

▶ Pump still running?

Yes:

Landing gear pump circuit breaker ..... **PULL (prevents pump burnout; hydraulic leak or pressure switch fault)**

Without pump pressure the gear is not locked up: expect the gear to sag out of the open wells and airspeed / performance to suffer. Keep below 140 KIAS.

Airspeed ..... **BELOW 140 KIAS**

Landing gear ..... **EXTEND with the hand pump (see Emergency Gear Extension) before landing**

Deferred: Hydraulic pump isolated – gear extended by hand pump, do not retract; hydraulic system inspection required. ..... **NOTED**

No:

Deferred: Hydraulic pump cycled abnormally – note for maintenance; check gear lights before landing. ..... **NOTED**

Gear Unsafe Indication / Gear Warning Horn

No gear-up or gear-down light after cycle, or horn sounds unexpectedly

Throttle ..... **NOTE – horn sounds when throttle is retarded below approx 12 in Hg with the gear not down and locked (also with flaps beyond approx 20° and gear up on some aircraft)**

Airspeed ..... **BELOW 140 KIAS**

Landing gear lever ..... **CHECK full travel into detent**

Landing gear pump circuit breaker ..... **CHECK IN / RESET once**

Gear lights ..... **PRESS TO TEST / swap bulbs if possible**

Landing gear lever ..... **RECYCLE once to the desired position (allow approx 10 sec; listen for the pump to stop)**

▶ Gear in desired position with correct light?

Yes:

Gear indication ..... **CONFIRM; horn silent at idle with gear down**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – will not retract:

Landing gear lever ..... **DOWN – leave down, green light**

Airspeed ..... **BELOW 140 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range, land as soon as practical. ..... **NOTED**

No – will not extend:

Use Emergency Gear Extension (Hand Pump).

Flap Failure / Asymmetry

Flap lever ..... **RETURN to the current position – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED (stay below 160 / 140 KIAS limits)**

No:

Landing ..... **AT CURRENT FLAP SETTING – flaps up approx 85–90 KIAS final, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Propeller Overspeed

RPM above 2,850 and not responding to prop control

Throttle ..... **REDUCE**

Propeller ..... **LOW RPM (full aft)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor is oil-operated)**

Power ..... **SET minimum that keeps RPM at or below 2,700**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

Cabin or Baggage Door Open in Flight

Aircraft	**CONTROLLABLE – fly the airplane**
Airspeed	**REDUCE to approx 100 KIAS**
Belts / loose items	**SECURE**
Cabin door	**OPEN STORM WINDOW / vents, then pull door closed and latch if it can be reached safely (co-pilot side only)**

Do not let a door distract from flying. A door that will not latch stays slightly open; expect noise and some buffet.

Deferred: Door unlatched – return and land as soon as practical, normal approach speed.	**NOTED**

Vacuum / Attitude Indicator Failure

Suction below 4.5 in Hg or gyro flags

Suction gauge	**CHECK**
Attitude / heading indicators	**CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**
Failed instruments	**COVER**
Autopilot	**OFF if coupled to the failed gyro**
IMC	**EXIT / AVOID**
Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach.	**NOTED**

Inadvertent Icing Encounter

Pitot heat	**ON**
Heading / altitude	**TURN BACK or CHANGE ALTITUDE to exit icing conditions**
Cabin heat / defroster	**FULL ON (windshield)**
Power	**INCREASE to minimise ice build-up on the propeller and to maintain airspeed; cycle RPM to shed prop ice**
Cowl flaps	**ADJUST to keep CHT in green (induction icing: expect power loss)**
Alternate static source	**ON if static ports suspect**

Stall speed rises with ice. Avoid steep turns; use flaps 0°–10° only and approach at 90–100 KIAS. Do not retract flaps once lowered if ice is on the tail.

Landing gear	**EXTEND EARLY (below 140 KIAS) – ice in the open gear wells may slow extension**
Deferred: Ice on airframe – flaps 0°–10°, approach 90–100 KIAS, longer landing distance, higher stall speed.	**NOTED**

Pitot / Static Source Blocked

Airspeed, altimeter or VSI erratic

Pitot heat	**ON**
Alternate static source	**ON (pull) – expect altimeter and airspeed to read differently; see POH calibration**
Cabin vents / windows	**CLOSED when using alternate static**
Attitude / power	**FLY KNOWN PITCH AND POWER (GPS ground speed as cross-check)**
Deferred: Pitot / static unreliable – fly known pitch and power on approach, add margin to approach speed.	**NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Throttle (memory)	**CLOSED**
Brakes (memory)	**AS REQUIRED – stop straight ahead**
Flaps	**UP**
Mixture	**IDLE CUTOFF**
Ignition switch	**OFF**
Fuel selector	**OFF**
Master switch	**OFF**

Engine Failure After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

Airspeed (memory)	**75–80 KIAS flaps down / 85 KIAS flaps up**

▶ Runway remaining to land on?

Yes:

Landing gear	**DOWN (if retracted and time permits – approx 10 sec cycle, no doors to wait for)**
Flaps	**AS REQUIRED**
Land	**STRAIGHT AHEAD**

No – land ahead off-field:

Landing gear	**DOWN on firm surface / UP on rough terrain or water**
Flaps	**30° when landing assured**
Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition switch	**OFF**
Master switch	**OFF (after gear and flaps set – both are electric)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

- Airspeed (memory) — **85 KIAS (best glide, gear & flaps up)**
- Propeller (memory) — **LOW RPM (full aft) to extend glide**
- Fuel selector (memory) — **SWITCH TANKS (feel for detent)**
- Auxiliary fuel pump (memory) — **ON (check fuel flow)**
- Mixture (memory) — **FULL RICH**
- Ignition switch (memory) — **BOTH (check)**

▶ Restart attempt practicable?

Yes – restart:

- Throttle — **1/2 OPEN**
- Starter — **ENGAGE if propeller has stopped**

▶ Engine restarted?

Yes:

- Power — **AS REQUIRED**
- Auxiliary fuel pump — **OFF once fuel flow is steady (ON if it fluctuates)**
- Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system or vapour. — **NOTED**

No:

- Auxiliary fuel pump — **OFF**

Continue with forced landing.

No – forced landing:

Forced landing without power:

- Seats / belts / harnesses — **SECURE**
- Field — **SELECT – plan to arrive high, flaps and gear only when assured**

▶ Surface firm and long enough for a gear-down landing?

Yes – firm surface:

- Landing gear — **DOWN below 140 KIAS (hand pump if hydraulic pump fails – allow time; no doors to open)**
- Gear-down light — **GREEN**

No – rough terrain, soft ground or water:

- Landing gear — **UP (leave retracted in the open wells)**

Gear-up touchdown: minimum speed, slightly nose high, wings level. Unlatch doors before touchdown.

- Flaps — **30° when landing assured**
- Airspeed — **75–80 KIAS on final (85 KIAS flaps up)**
- Mixture — **IDLE CUTOFF**
- Fuel selector — **OFF**
- Ignition switch — **OFF**
- Master switch — **OFF (after gear and flaps set)**
- Cabin doors — **UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

- Mixture (memory) — **IDLE CUTOFF**
- Fuel selector (memory) — **OFF**
- Auxiliary fuel pump (memory) — **OFF**
- Landing gear — **DOWN now if a gear-down landing is planned (below 140 KIAS – electric pump needs the master ON)**
- Master switch — **OFF**
- Cabin heat / cabin air — **OFF (overhead vents open as required)**
- Airspeed — **100 KIAS – if fire persists, increase speed to find one that gives an incombustible mixture**
- Forced landing — **EXECUTE (do not attempt restart)**

With the master OFF the gear can only be extended with the hand pump and the flaps will not move.

Engine Fire During Start

Starter (memory) ..... **CONTINUE CRANKING to draw fire into engine**

▶ **Engine starts?**

Yes:

- Throttle ..... **1,700 RPM for a few minutes**
- Engine ..... **SHUT DOWN and inspect for damage**

No:

- Mixture ..... **IDLE CUTOFF**
- Throttle ..... **FULL OPEN**
- Auxiliary fuel pump ..... **OFF**
- Fuel selector ..... **OFF**
- Starter ..... **CONTINUE CRANKING**
- Ignition switch ..... **OFF**
- Master switch ..... **OFF**
- Fire extinguisher ..... **OBTAIN / USE**
- Aircraft ..... **EVACUATE**

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

Master switch (memory) ..... **OFF**

Avionics power switch (memory) ..... **OFF**

All other switches (except ignition) (memory) ..... **OFF**

Cabin heat / vents ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE (then ventilate cabin once fire is out)**

▶ **Fire out and electrical power required?**

Yes:

- Circuit breakers ..... **CHECK – do not reset a faulted breaker**
- Master switch ..... **ON**
- Avionics power switch / electrical equipment ..... **ON one at a time, pausing to check for smoke**
- Landing gear ..... **EXTEND EARLY while power is available**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Gear pump and flaps are electric: plan a hand-pump gear extension (see Emergency Gear Extension) and a flaps-up landing at 85–90 KIAS.

- Deferred: Electrical power off – hand-pump gear extension, flaps-up landing, no landing light or radio. ..... **NOTED**

Emergency Descent

Throttle (memory) ..... **CLOSED**

Propeller (memory) ..... **HIGH RPM**

Landing gear (memory) ..... **DOWN (below 140 KIAS)**

Flaps (memory) ..... **UP**

Airspeed (memory) ..... **165 KIAS gear down (Vle); Va approx 133 KIAS in turbulence**

Mixture ..... **ENRICH as required**

Cowl flaps ..... **CLOSED**

Emergency Gear Extension (Hand Pump)

Gear does not extend hydraulically or no green light

Airspeed	**BELOW 140 KIAS (slower reduces pump effort – approx 100 KIAS)**
Master switch	**ON**
Landing gear lever	**DOWN**
Landing gear pump circuit breaker	**PULL if the pump runs continuously or has tripped (otherwise leave in)**
Emergency hand pump	**EXTEND HANDLE (floor between front seats) and PUMP until the green light comes on and the handle becomes firm (no main gear doors – gear extends straight into the airstream; typically 40–60 strokes)**
Gear-down light	**GREEN**
Hand pump handle	**STOW**

Do not retract the gear or reset the pump breaker in flight after a hand-pump extension. Leave the gear down.

▶ Green light obtained?

Yes:

Deferred: Gear extended with the hand pump – do not retract; gear and hydraulic system inspection required. ..... **NOTED**

No:

Gear-down light bulb	**CHECK (swap with gear-up bulb if possible); gear horn silent at idle?**

If still unsafe, continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, exits, both doors)**

▶ Gear position?

All up:

Landing gear lever	**UP – do not cycle**

Nose gear only down:

Landing gear	**LEAVE AS IS – land on nose gear and belly, hold nose up as long as practical**

One main gear up:

Landing gear	**RETRACT ALL if possible (landing on one main is worse than gear up); otherwise land on the extended gear side of the runway, wing low to that side**
Flaps	**30°**
Approach	**NORMAL, 75–80 KIAS**
Auxiliary fuel pump	**OFF**
Cabin doors	**UNLATCHED**

On short final:

Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition switch	**OFF**
Master switch	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**

Inadvertent Spin

Throttle (memory)	**IDLE**
Ailerons (memory)	**NEUTRAL**
Rudder (memory)	**FULL OPPOSITE to rotation**
Elevator (memory)	**BRISKLY FORWARD to break stall (just past neutral)**
Flaps	**UP (if extended)**
Rudder	**NEUTRAL when rotation stops**
Recovery	**SMOOTH PULL-OUT from the dive – respect Vne 204 KIAS and gear / flap speeds**

Intentional spins are prohibited. Lower nose first if a secondary stall threatens.